# DFIP: A Systems Study of Fidelity-Aware Inference for Open Decision Models

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## Abstract

Open decision models return categorical distributions, Boolean confidences, or ordinal scores, but their inference paths can retain substantial host dispatch and full-vocabulary projection. We study a decision-focused inference path (DFIP) that preserves released prompt and readout contracts while applying selective CUDA graph replay and candidate-token projection. Pinned checkpoints from Liquid AI, Cloudflare, Perplexity, Superagent, and Supersonic Labs run on Modal GPUs. We separate randomized paired resident measurements, component ablations, diagnostic workloads derived from BANKING77, optimized-native comparisons, and external open-loop serving tests. With optional native Qwen kernels installed, retained short Boolean runs yield 2.47× for PPLX-Decider over five containers and 1.45× for Clef over four of five planned containers; Clef's fifth fails in native preflight. Against Liquid's released compiler setting, DFIP is effectively tied on the Boolean workload and has no general latency advantage across three short workloads. Graph replay supplies most of the initial gain, while candidate projection alone does not establish a general speedup. Direct audits of 3,024 decisions preserve returned choices for Decider and Clef, but Clef changes scores and probabilities; matched serving traffic also crosses sampled returned-confidence thresholds. Sustained changing-request traffic exposes queueing, overload losses, and replay-path CUDA errors beyond resident-call measurements. Concurrent dispatch to native SGLang sustains nearly 30 requests/s in the tested Security-One configuration without drops. We release per-call samples, request accounting, failure traces, pinned revisions, and source snapshots. The contribution is a bounded empirical characterization of established techniques and their fidelity and serving limits, rather than a new graph algorithm or production readiness certification.

## 1 Introduction

Routing, policy classification, and structured scoring often require a small answer set. In these applications, inference returns a decision distribution rather than an open-ended continuation. A deployment can therefore spend much of its request time on tokenization, Python dispatch, GPU launch overhead, and output computation whose results are discarded. Reducing this work is attractive for both latency and compute cost, but the relevant contract includes calibrated probabilities and ordinal scores as well as the winning label.

Existing serving systems already support structured programs and constrained selection. SGLang, for example, includes a selection primitive and exploits execution structure in its runtime [15]. CUDA graphs and compiler transformations likewise provide established tools for reducing launch overhead [3, 9]. DFIP does not introduce these ideas. Our question is how far a small, model-aware implementation can apply them to recently released open decision checkpoints without silently changing their interfaces or numerical behavior.

We ask four research questions. **RQ1:** Which released inference paths benefit from selective replay relative to stronger native baselines across short, long, batched, and multi-question workloads? **RQ2:** How much improvement comes from graph replay versus candidate projection? **RQ3:** Which optimizations preserve decision and probability behavior, and where do they fail? **RQ4:** Do resident-call gains survive changing external arrivals? Our contribution is a reproducible implementation and empirical characterization with negative results, rather than a universal acceleration claim.

The artifact is available for author review at `https://github.com/superagent-ai/dfip`. At this draft stage the repository is private; public artifact access and a frozen release must be resolved before submission. The implementation uses Python for adapters, validation, and orchestration, and Triton [14] for experimental GPU kernels. All checkpoints and benchmark datasets remain on Modal; a lightweight local client sends structured requests.

## 2 Decision Contracts and Scope

An input contains a state $s$ and a set of typed questions $Q$. A choice question returns probabilities over a finite

set of criteria; a Boolean question returns a confidence in the affirmative answer; an ordinal question returns a distribution over ordered levels and may expose its expected score. The checkpoints implement these contracts using different prompt encodings and different heads. A common client interface does not imply interchangeable internal computation.

Julia-1 uses a ModernBERT-based encoder and marker-oriented decision readout. Liquid's d1-3B uses its released LFM-based engine, including a packed trunk/branch path for multiple questions. Clef-Flash uses a Qwen backbone and a joint schema head. PPLX-Decider uses its released Qwen-based decision model. Security-One is evaluated through its SGLang path [2, 6, 8, 12, 13]. We retain each released prompt construction, candidate ordering, alias handling, temperature, and answer conversion. We do not train, quantize, or distill the checkpoints.

These are five selected checkpoints, not an exhaustive sample of the Hugging Face decision-model tag. Text-only workloads are tested even when a checkpoint supports images. Tests do not establish safety, deployment suitability, or superior task quality. We also do not compare with generic one-token generation as though it were a semantically identical baseline.

### 2.1 When candidate projection is valid

Let $h \in \mathbb{R}^d$ be the hidden vector at a released answer position and $z = Wh$ the vocabulary logits. For a native readout that groups answer-token aliases $A_c$, a representative contract is

$$r_c = \max_{t \in A_c} z_t, \qquad p_c = \frac{\exp(r_c/\tau)}{\sum_j \exp(r_j/\tau)}. \tag{1}$$

If $U = \bigcup_c A_c$, computing $W_U h$ preserves these readout logits in real arithmetic. A common log-normalization constant cancels when the selected scores are subsequently normalized over candidates. The condition is that the released readout consumes only the covered token groups: arbitrary generation, a full-vocabulary confidence, top-$k$ operations over the entire vocabulary, or an incomplete alias union do not satisfy it. Other decision heads need their own contract analysis.

This identity is not a finite-precision equivalence theorem. A smaller matrix multiply can choose a different reduction order or precision path. If $\|p - \hat{p}\|_\infty \le \epsilon$, an argmax with a margin greater than $2\epsilon$ is preserved, while a threshold comparison at $a$ is preserved only when $|p_c - a| > \epsilon$. For ordinal levels $v_c$, expected-score drift is bounded by $\sum_c |v_c|\,|p_c - \hat{p}_c|$. These bounds concern the probabilities actually consumed by the readout; separately rounded API fields require direct checks. Agreement of the winning label alone is thus insufficient for probability-sensitive applications.

## 3 Implementation

### 3.1 Adapters and selective graph replay

The runtime loads an immutable checkpoint revision and uses an adapter to invoke the released inference engine. DFIP wraps a compatible forward boundary with static input buffers and a shape cache. A key records tensor dimensions, strides, dtype, device, and supported scalar metadata. Replays copy current inputs into the captured buffers; the cache stores computation, never a previous prediction. Calls sharing mutable buffers are serialized in the serving engine.

The v1 tested policy admits a shape after three encounters and bounds the graph cache to four entries. Liquid, Clef, and PPLX-Decider use a 512-token capture limit. Clef and PPLX-Decider limit capture to a single forward row. Unsupported paths use the released native implementation. PPLX-Decider's hybrid recurrence path additionally keys valid lengths, since shape equality alone cannot prove control-flow equivalence. For Julia, an equivalent local attention mask is constructed on the GPU to make capture possible. These adaptations are implementation choices rather than claims of novel graph scheduling.

Captured allocation lifetime is another constraint. The first expanded protocol encountered illegal GPU memory access in three checkpoint applications. The error was observed during benchmark execution, sometimes after a mode transition; asynchronous reporting does not identify the responsible kernel. Failure excerpts are retained. A follow-up implementation synchronizes before clearing or evicting a graph. A 60-call synthetic-kernel test covering shape eviction and cache clearing passes, but Julia still fails during the new-workload fidelity run after this change. This does not resolve the model-level failure, and we do not pool the follow-up with the original protocol. After the strengthened serving study also exposes Decider replay errors and recapture overhead, the review branch defaults all five checkpoints to native inference. Experimental replay and head paths remain available through research entry points; Security-One retains its native SGLang substrate. Existing deployed applications were not changed.

### 3.2 Candidate heads and rejected kernels

For d1-3B, the adapter derives answer-token groups using the released tokenizer and readout code. A head projects the verified union with either a Triton reduction kernel or PyTorch's linear operation. New groups extend that union and invalidate existing captures. To preserve the released engine's tensor interface, the current prototype reconstructs a vocabulary-shaped tensor with uncomputed positions set to negative infinity. It therefore removes much projection arithmetic but retains output allocation, scatter, and readout overhead. This is an explicit limitation rather than a minimal-width output implementation.

Security-One's experimental SGLang patch projects its

verified answer-code rows at the native output boundary. It is isolated from the selected deployment policy, which remains native SGLang. The comparison uses both Triton and cuBLAS-backed projection. Experimental RMSNorm replacements and released compiler options from the earlier development study are retained as supplementary evidence. They are not enabled merely because a local kernel test passes: full-model probability and decision checks determine whether a path is admissible.

### 3.3 Modal execution and packaging

GPU workers use pinned model revisions and persistent remote volumes. The local package depends on the Modal client, not PyTorch or checkpoint weights. It exposes preparation, deployment, inference, and validation commands, while accelerator dependencies live in remote images. Deployment can scale to zero. The study initially ran model applications concurrently and reached the workspace's GPU concurrency limit; subsequent workers are capped at one container, and follow-up experiments are serialized. Concurrency changes how quickly experiments finish and is not a source of model acceleration.

## 4 Experimental Method

### 4.1 Hardware, software, and baselines

Julia, d1-3B, and Clef run on one NVIDIA L40S each; PPLX-Decider and Security-One run on one NVIDIA B200 each. Worker requests use four CPU cores and 32 GiB of host memory for the smaller models, and eight cores and 96 GiB for the larger models. Per-round GPU identity, driver, CUDA version, core package versions, and container identity accompany the raw samples. Separate CPU-only inventories list all installed packages in the v3 image definitions; they are supplementary environment evidence rather than additional GPU rounds. The Transformers reference images use PyTorch 2.14.0, CUDA 13.0, Triton 3.8.0, and Transformers 5.17.0; Julia uses Transformers 5.0.0. Security-One uses the SGLang 0.5.19 CUDA 13 image.

The primary baseline is the released eager/native adapter in the same environment. This is not a comparison against every optimized serving engine. In particular, the initial Qwen reference environment lacks the optional `causal-conv1d` extension and uses a PyTorch fallback. An optional extension build is recorded separately; results from such a build must not be mixed with the fallback baseline. Consequently, gains for Clef and PPLX-Decider cannot be interpreted as improvements over an optimized-kernel ceiling. SGLang already supplies an optimized serving substrate for Security-One.

### 4.2 Workloads and fidelity sets

Nine timing workloads cover a short Boolean decision, a short categorical decision, three questions on one state, medium and long repeated-log contexts, candidate widths 2, 8, and 16, and eight requests in one batch. The exact states and schemas are defined in the artifact. Contexts are synthetic; token counts differ across released prompt encodings. A 120-call trace changes state content and lengths across 40 context sizes to test admission and replay without prediction caching.

The first fidelity set contains 400 typed-decision cases with 2,000 questions from the pinned `LocalLLaMA/typed-decisions` test split. This set was used during earlier DFIP development and is not a pristine holdout. A second workload samples 256 test utterances from BANKING77 [1]. For each utterance, we construct one single-choice record and one record containing choice, Boolean, and ordinal questions: 512 records and 1,024 decisions in total. Choice candidate widths cycle through 2, 4, 8, and 16 and include the gold intent. Boolean targets alternate between matching and nonmatching intents. Derived ordinal questions are diagnostic and have no validated annotation. Dataset revision hashes, content checksums, seeds, and generated-record hashes are included. Checkpoint training overlap with either corpus is unknown.

We compare complete returned probability distributions, their argmax labels, probability-derived expected ordinal scores, and probability-slot threshold crossings at 0.1, 0.25, 0.5, 0.75, and 0.9. Clef rounds probability slots after selecting a categorical answer and computes its returned score from unrounded probabilities. The original v1/v3 distribution reports therefore do not independently establish literal answer-field equality at rounded ties. Separate instrumented v3 audits compare returned choice, score, and confidence fields directly on all 3,024 decisions; later serving records also retain these fields. Measured probability drift concerns the released interface rather than unrounded internal logits. Diagnostic accuracy, Brier score, and equal-width expected calibration error are available for the derived choice/Boolean labels, but repeated single-choice and multi-question views of one utterance are correlated. They are not independent quality examples.

### 4.3 Timing and statistical unit

The protocol plans five independent single-use container rounds per checkpoint, with 50 timed calls per workload and mode after four admission/warmup calls. Mode and workload orders are randomized by round. GPU synchronization brackets the resident Transformers API timing, which includes encoding, forward computation, and native readout. Security-One timing includes the in-container HTTP client and SGLang scheduler. It excludes external client networking and server startup. Its native runs

bracket the randomized head experiments to expose drift across server restarts.

The statistical unit is a container round. We compute each workload's median within each round, pair native and tested-policy medians, and report their geometric-mean ratio. Intervals are 95% percentile bootstrap intervals over paired log ratios with 10,000 resamples; the 50 calls are not treated as 50 independent deployment replications. Small samples and cloud variability remain limitations. For Security-One, the within-round reference is the geometric mean of the two native medians.

Julia's initial application returned no complete expanded-protocol round. Liquid and Clef each returned three rounds before an application-level GPU error aborted the remaining work. PPLX-Decider and Security-One completed five rounds each. Security-One's last container also emitted driver graphics-exception warnings before the first measured mode; the function continued and returned a complete round. We retain these warnings without assigning a kernel root cause. Tables are generated from complete retained JSON records and report actual counts. Attrition can be nonrandom, so intervals based on retained runs are conditional and may be optimistic. We do not impute failed rounds or substitute old one-container samples as independent repetitions.

### 4.4 Strengthened native and serving protocol

We freeze a separate v3 protocol rather than pool new measurements with v1. For Clef and PPLX-Decider, five single-use container rounds are planned with the same nine workloads, randomized native/DFIP order, and 50 warm samples. The image adds `causal-conv1d` 1.7.0 to `fla-core` 0.5.2 for the released Qwen linear-attention path [4]. SDPA and PPLX-Decider's noncausal attention semantics remain intact. This is a stronger same-image native comparison, not a claim to equal every specialized backend; differences between the old and new images cannot isolate the causal effect of extension installation. A separate three-round Liquid experiment compares native eager, the released runner's `torch.compile(mode="reduce-overhead")` setting [10], and DFIP on three short workloads. It checks 128 single categorical requests with changing content, length, and candidate width; this compiler fidelity check does not cover every Boolean or ordinal timing schema. Compiler admission and failed modes are retained.

The serving study plans three independent application/container rounds for Clef, PPLX-Decider, and selected native Security-One. The first two compare native with experimental graph replay in randomized mode order within one worker. An external author workstation with Swedish network egress sends authenticated Modal RPC requests at fixed open-loop rates of 5, 15, and 30 requests/s for 60 seconds each. Contents, twelve recurring state-repetition counts, and candidate widths 2/8/16 change deterministically. This diversity stress case can exceed the four-entry graph cache; it does not estimate steady-state serving for a single captured shape. The payloads also differ from the resident short Boolean workload, so serving throughput is not divided by its latency to infer a workload-independent gain. Latency starts at scheduled arrival, includes client dispatch delay, transport, and queueing, and ends when the result returns. Each phase drains fully before the next. Worker initialization and 48 admission calls are recorded separately. Checkpoint and compiler artifacts persist in the remote volume; initialization is not an empty-cache download or compilation benchmark.

The worker accepts up to 64 concurrent RPC inputs. A lock serializes Transformer GPU computation to protect mutable capture buffers. Security-One separately compares this serialized client dispatch with concurrent requests to the native SGLang scheduler, which owns its GPU state and can batch requests. Existing engine settings remain fixed, including 32 maximum running requests, disabled radix caching, and SGLang's own CUDA graph execution. Native SGLang therefore already includes backend optimizations; the comparison changes wrapper dispatch rather than tuning the entire backend. In that mode the reported wrapper queue is zero by construction; service time includes the internal HTTP request and scheduler queue, which are not separated by this instrumentation. The client permits 128 requests in flight; excess arrivals are counted as dropped. Errors and drops stay in the offered-load denominator. Tail latency is conditional on successful requests and is reported alongside loss and within-window completion throughput. We aggregate per-container quantiles, not thousands of calls as independent hardware replications. Before/after checks use 128 matched requests; returned stream distributions enable additional ID-matched comparisons across modes, with unmatched requests counted separately. These are minute-scale tests of these wrappers, not a tuned dynamic-batching throughput ceiling or hours-long production reliability certification.

## 5 Results

### 5.1 RQ1: Comparison with stronger native baselines

Table 1 reports the repeated v3 comparison in the image with native CUDA convolution and FLA bindings. PPLX-Decider completes five planned rounds; Clef completes four, with its fifth container failing during native reference collection before any timed mode or custom graph admission. The asynchronous error does not identify a kernel root cause and shows that failure evidence is not confined to custom graph replay. Clef's intervals remain conditional on retained rounds. Estimates are paired

within the new image and are not pooled with Table 6. In particular, a difference between the two native columns across experiments is not an isolated extension-installation effect: container hardware, allocator history, and image dependencies also differ.

Figure 1 extends the v3 comparison across all nine workloads. Gains are concentrated in repeated short single-decision requests. Long contexts and batches use native fallback, and PPLX-Decider's multi-question path also falls back. The finite cache and admission policy are part of the tested system, so a model-wide speedup would hide substantial workload dependence. Container variability remains large: PPLX-Decider's native short Boolean medians range from 47.22 to 114.56 ms. Pairing avoids dividing unrelated runs, but does not eliminate mode-order, clock, or allocator-history effects.

All three Liquid compiler containers complete. Table 2 shows no short Boolean advantage over the released compiler setting: the paired ratio is 1.01× with an interval spanning one. DFIP is slightly slower on the categorical workload and slightly faster on three questions. Across the same 128 changing categorical requests in each round, both paths preserve returned choices and sampled threshold classifications, but compilation reaches maximum probability drift 0.0220 versus $7.67 \times 10^{-5}$ for DFIP relative to eager native. This is a latency–fidelity tradeoff on a small diagnostic set, not evidence of degraded task accuracy. The compiler emits a warning about accumulating captures for distinct input sizes; the artifact retains it and admission cost. We do not infer a fully tuned compiler ceiling or exact numerical equivalence.

The initial v1 study uses a weaker Qwen fallback image and retains only three Liquid and Clef rounds, alongside five Decider and Security-One rounds. Its complete resident results are in Appendix D. These retain useful ablation and failure evidence but are not pooled into the optimized-native estimates. Security-One's head experiment yields no convincing short Boolean improvement over its optimized SGLang substrate; the default remains native SGLang.

## 5.2 RQ2: Graph replay versus projection

Table 3 isolates the d1-3B components on the short Boolean workload. Graph-only replay supplies the large part of the observed reduction. Candidate projection without replay does not establish a general gain, and changing between cuBLAS and Triton within the graphed path gives a much smaller difference. This is consistent with a request whose total time also contains encoding, backbone execution, and Python readout.

This ablation supports retaining a model-aware replay path but does not support marketing DFIP as a generally faster custom matrix-multiply kernel. The prototype still reconstructs a vocabulary-shaped output. A narrower readout interface remains an engineering question; the v3 experiment tests a stronger native baseline separately.

## 5.3 RQ3: Decision agreement is weaker than fidelity

Appendix Table 7 combines the two fidelity sets for complete v1 round-zero results. No tested policy in the table changes an argmax reconstructed from returned probabilities or crosses the five sampled probability thresholds. Nevertheless, d1-3B and Clef exhibit measurable drift on the new workload. The absence of crossings at these five thresholds does not establish safety for arbitrary thresholds close to a model confidence, and exact real-arithmetic projection does not imply bitwise equality.

The Security-One cuBLAS-backed candidate head provides a clear counterexample to using label agreement as the only gate: it preserves all 2,000 returned-distribution argmax labels in the typed set but reaches a maximum probability difference of approximately 0.0393. The Triton variant is much closer to the native probabilities, with a maximum difference below $2 \times 10^{-7}$ on that set. Neither result proves that candidate projection yields a worthwhile full-request speedup on the SGLang substrate. The selected deployment therefore retains native SGLang. A separate single-container Clef follow-up with the optional native convolution extension also preserves returned-distribution argmax labels, but produces two sampled threshold crossings and a maximum new-workload probability difference of 0.0129 (Appendix E). This is a concrete probability-sensitive failure even when argmax agreement is complete.

Table 4 reports the separate literal-readout audits, each evaluating 3,024 decisions. PPLX-Decider has zero probability drift, zero changes in 1,112 returned choice fields, and zero drift in 1,056 score and 2,168 confidence fields on this sample. Clef also preserves all 1,112 returned choices, but has maximum returned score drift 0.0188 and confidence drift 0.0129. Its probability-slot crossings at 0.1 and 0.25 reproduce the earlier negative result. These are single-container audits with fixed native-then-DFIP order, not additional independent timing rounds. In particular, the observed Clef probability crossings are not demonstrated Boolean accept/deny flips.

Earlier exploratory RMSNorm substitutions changed winning labels for Liquid, Clef, and PPLX-Decider, and a Julia compiler experiment changed decisions as well. Those observations motivated the full-model gates and are archived separately from the repeated-run protocol. Kernel-level closeness to a reference reduction is insufficient: changed accumulation and rounding can propagate through many layers or a calibrated readout. The original expanded run's GPU failures similarly prevent a blanket reliability claim about all replay shapes.

| Checkpoint | Rounds | Native (ms) | DFIP (ms) | Speedup [95% CI] |
|---|---|---|---|---|
| Clef-Flash | 4/5 | 51.69 | 35.68 | 1.45 [1.33, 1.71] |
| PPLX-Decider | 5/5 | 74.54 | 30.14 | 2.47 [1.88, 3.26] |

Table 1: Separate v3 short Boolean comparison with optional native CUDA kernels installed. Latencies are geometric means of within-container medians. Counts are completed/planned container rounds; intervals resample paired container medians. This tests released native decision semantics, not generic causal generation or a tuned serving-engine ceiling.

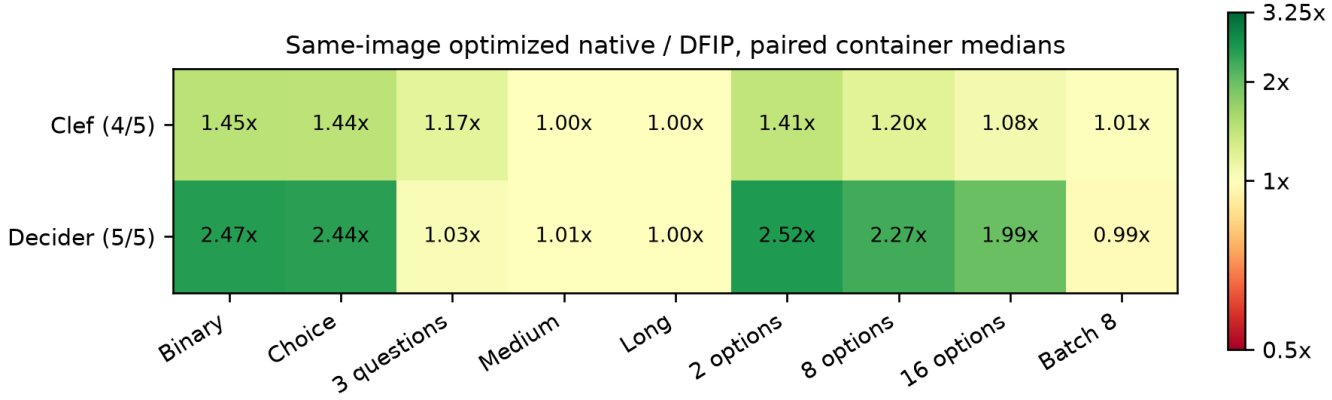



Figure 1: V3 paired resident-API speedup with optional native kernels installed. Clef estimates condition on four completed rounds; PPLX-Decider completes five. Values below one indicate a slowdown. Per-workload conditional intervals and raw samples are in the artifact.

### 5.4 RQ4: External arrivals and serving limits

Figure 2 summarizes serving behavior; Table 5 in the appendix reports observed phases, including completed phases from containers that later fail. Counts differ when a failure prevents later rates or modes; omitted phases are not successful zero-loss measurements. Tail quantiles concern successful requests, while the loss denominator contains all offered arrivals. Container-level tail ranges are retained in the artifact because cloud variability is large.

All three PPLX-Decider attempts are retained: two complete and one fails with CUDA illegal-memory-access errors during its first graph traffic phase, after completing all native phases. At 5 offered requests/s, native has zero loss with a geometric-mean p95 of 0.81 s. The graph path yields only 2.04 within-window completions/s, 23.0% loss, and a successful-request p95 of 49.41 s (container range 21.79–76.55 s). At 30 offered requests/s, the two observed graph phases have 83.1% loss. The two complete graph rounds preserve returned choices and probabilities on matched traffic; all three together provide only 1,761 matched successful request pairs across rounds, with repeated trace inputs. This selected subset cannot certify omitted or failed requests.

The diagnostics expose frequent recapture: the second Decider attempt, which completes, records 820 captures and 915 replays, including admission and fidelity checks, with most replays therefore occurring in the same call as capture. At 30 offered requests/s its graph service median is 524 ms versus 129 ms for native. This supports recapture overhead as an explanation for the performance reversal on this diversity stress case, but does not localize the asynchronous CUDA fault or establish performance for a single recurring shape. It rules out extrapolating the resident 2.47× gain into a general serving claim. The review package consequently defaults Decider to native, with replay available only in research entry points; existing deployments are unchanged.

All three Clef serving rounds complete without request errors. At 5 offered requests/s, native has zero loss and p95 0.34 s; replay has 4.9% loss, 2.68 within-window completions/s, and p95 39.44 s. At 30 offered requests/s, replay loses 83.1% of arrivals versus 54.3% for the serialized native wrapper. No returned choices change among 2,473 matched successful request pairs across rounds, but maximum probability and returned-confidence drift is 0.0056. Twenty-one of these pairs cross the literal returned-confidence threshold 0.9, with repeated trace inputs across rounds. This differs from the earlier ordinal probability-slot crossings: it directly changes a hypothetical confidence-threshold routing policy, without demonstrating worse task accuracy or a Boolean answer flip. The native-preflight failure in the separate optimized resident experiment remains a distinct reliability outcome.

| Workload | Rounds | Compiled (ms) | DFIP (ms) | Ratio [95% CI] |
|---|---|---|---|---|
| short binary | 3/3 | 10.74 | 10.66 | 1.01 [0.97, 1.09] |
| short choice | 3/3 | 10.47 | 10.97 | 0.95 [0.95, 0.96] |
| short 3 questions | 3/3 | 21.78 | 20.84 | 1.05 [1.03, 1.07] |

Table 2: Liquid v3 comparison against the released `reduce-overhead` compiler setting, three completed/planned container rounds. The ratio is compiled-native latency divided by DFIP latency; below one favors compilation. Intervals resample paired container medians.

| d1-3B mode | Median (ms) | Speedup [95% CI] |
|---|---|---|
| `native` | 18.69 | 1.00 [1.00, 1.00] |
| `graphs_only` | 11.47 | 1.63 [1.18, 1.92] |
| `candidate_cublas` | 18.40 | 1.02 [0.95, 1.07] |
| `candidate_triton` | 18.12 | 1.03 [1.00, 1.06] |
| `production_cublas` | 10.69 | 1.75 [1.26, 2.08] |
| `production` | 10.68 | 1.75 [1.26, 2.07] |

Table 3: Short Boolean ablation, three retained paired rounds. `production` combines graph replay and the Triton candidate head. The dynamically extended union contains 40–42 tokens during these timing observations; that order-dependent detail limits a strict fixed-width kernel interpretation.

| Checkpoint | Choice changes | Max $\lvert\Delta p\rvert$ | Max $\lvert\Delta s\rvert$ |
|---|---|---|---|
| Clef-Flash | 0/1112 | 0.0129 | 0.0188 |
| PPLX-Decider | 0/1112 | 0.0000 | 0.0000 |

Table 4: V3 direct answer audits. Choice changes use literal returned fields, with 1,112 categorical choices compared per model; maximum score drift uses 1,056 returned ordinal scores. Probability drift covers all 3,024 decisions. Fixed native-then-DFIP order, one container per model; no all-input guarantee.

All three Security-One rounds complete. Concurrent dispatch to the same native SGLang engine sustains 29.76 within-window completions/s at 30 offered requests/s, with no drops or request errors and geometric-mean p95 0.65 s (range 0.55–0.71 s). Serialized dispatch gives 18.33 completions/s, 31.8% drops, and p95 6.95 s. The benefit is workload dependent: at 15 requests/s, concurrent p95 ranges from 0.43 to 3.56 s and its geometric mean is slightly higher than serialized dispatch. All 7,283 matched successful request pairs across rounds preserve literal choices; maximum probability drift is $1.56 \times 10^{-15}$, with no sampled crossings. This is a wrapper-dispatch result using an already optimized native engine, not evidence for a custom output-head kernel or a uniform latency improvement. Its minute-scale scope and selected matched subset remain limitations.

# 6 Cost Interpretation and Deployment Limits

At Modal's listed rates on October 12, 2026, an L40S costs \$0.000542 per GPU-second and a B200 costs \$0.001736 [7]. A narrow compute estimate for $D$ decisions per call is

$$C_{1000}^{\mathrm{GPU}} = 1000\, r_{\mathrm{GPU}}\, T_{\mathrm{call}}/D, \tag{2}$$

where $T_{\mathrm{call}}$ is in seconds. The artifact computes this quantity from the retained medians. It is not a measured invoice or a cost-per-request claim for a network service. CPU, memory, model loading, graph admission, idle container lifetime, storage, networking, and plan fees are excluded.

For $N$ calls within one billed container lifetime, a more useful approximation is

$$C \approx r_{\mathrm{resources}}(T_{\mathrm{load}} + T_{\mathrm{admit}} + NT_{\mathrm{serve}} + T_{\mathrm{idle}}). \tag{3}$$

Modal bills load and processing time and the configured idle window [7]. For sparse traffic these terms can dominate a small kernel reduction. Capping benchmark concurrency avoids a plan upgrade but does not make a fixed amount of GPU work free. The v3 serving protocol adds external transport, fixed arrivals, queueing, and tail-latency measurements to the resident study. It still does not establish production traffic representativeness, measured billing savings, or invoice reconciliation. A resource-rate estimate must not be presented as an observed bill reduction.

# 7 Related Work

SGLang [15] combines structured language-model programs with a runtime that reuses execution structure. DFIP uses a narrower native decision contract and does not claim structured selection as a new primitive. PagedAttention and vLLM [5] address KV-cache memory management and serving throughput for generation; those mechanisms and their arrival-rate evaluation are distinct from this serial decision-path study.

PyGraph [3] develops compiler support for robust CUDA graph use and evaluates component contributions across workloads. Its broader treatment makes clear that a small shape cache is not a novel graph compiler. PyTorch's CUDA graph documentation [9] also specifies

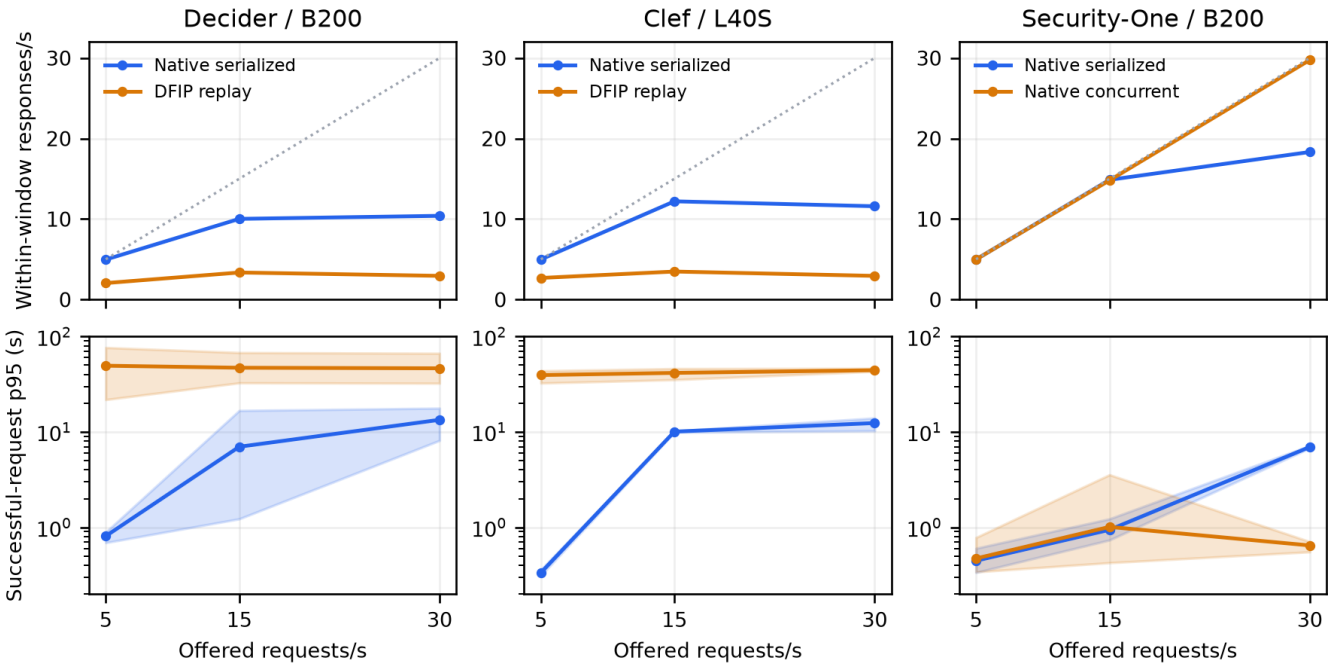



Figure 2: V3 external arrivals: mean within-window completion throughput (top) and geometric-mean successful-request p95 (bottom). Dotted lines mark offered load. Shading shows observed container p95 ranges, not confidence intervals. Decider/Clef compare native with experimental DFIP replay; Security-One compares dispatch to the same native SGLang engine. Counts and overload loss appear in Table 5; incomplete rounds contribute only observed phases.

static-address and control-flow constraints. Triton [14] provides a tiled programming model for custom GPU computation; DFIP applies it to a small candidate projection rather than proposing a new language or tensor primitive.

MLPerf Inference [11] emphasizes workload scenarios, quality constraints, and latency/throughput measurement. This informs our separation of fidelity from timing, but the present study is not MLPerf-compliant and does not measure all of its serving scenarios. BANKING77 [1] supplies an independently sourced intent workload; deriving a finite candidate set from its labels is our diagnostic construction, not the original benchmark task or a new validated scoring dataset.

# 8 Threats to Validity and Open Work

**Baseline strength.** Optional native convolution kernels are absent in the initial Transformers environment. The separate v3 comparisons add native CUDA kernels and a released compiler setting, but do not exhaust tuned serving backends. **Selection and attrition.** Five checkpoints are chosen for interface diversity. Failed rounds are not missing at random, and three-round conditional intervals provide weak evidence of population stability. Julia lacks a complete initial expanded-protocol result.

**Workload validity.** Timing states are synthetic; the serving study uses one external client geography and three fixed arrival rates over minute-scale windows. It cannot establish hours-long reliability, burst resilience, autoscaling behavior, or a Transformer dynamic-batching ceiling. BANKING77 schemas are derived, and duplicated utterance views are correlated. The typed set was used during development, and checkpoint training overlap is unknown. Image paths, adversarial states, maximum supported contexts, and multi-GPU configurations are untested here.

**Numerical validity.** Returned probabilities may be rounded, calibration magnifies some logit differences, and five threshold values cannot cover all application policies. No observed label flip is a finite sample statement. Observed drift is not localized to one kernel; capture compatibility and arithmetic differences may both matter. Dynamic candidate coverage and graph admission interact with workload order. The initial failure's asynchronous traceback cannot establish a kernel-level root cause, and a conservative synchronization change requires independent validation.

**Reproducibility.** Model and dataset revisions and package versions are recorded, but cloud scheduling, server placement, hardware contention, driver updates, transitive dependencies, and graph-library behavior are not fully controlled. Fresh containers do not guarantee independent physical hosts. The artifact preserves source versions and failed runs. Before an arXiv submission, the authors should review the failures and numerical differences, freeze a publicly accessible artifact, and check every generated table against the archived results. Cloud experiments and

a manuscript draft do not replace author review.

## 9 Conclusion

DFIP shows how a lightweight, model-aware decision inference implementation can combine native semantics with established replay and candidate-readout techniques. The retained results favor selective graph replay for repeated short workloads and provide weaker evidence for custom output projection as a full-request optimization. Probability drift, GPU failures, and recapture overhead under diverse arrivals define meaningful boundaries of the current prototype. A warm isolated-call gain does not by itself justify enabling replay for changing serving traffic. A useful research contribution is therefore the characterization and reproducible evidence, including rejected paths, rather than an unconditional promise of faster or exact inference across decision models.

## AI Assistance and Author Review

Codex, an AI coding assistant, assisted with implementation, experiment execution, result inspection, literature searches, and manuscript drafting. Numerical tables are generated from archived measurements rather than generated estimates. This review draft requires verification of the experiments, references, and interpretations by the named authors before submission; the authors retain responsibility for the paper's contents.

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# A Artifact and Reproduction

The repository contains a pinned model registry, remote preparation code, research entry points, raw per-call timing JSON, derived-workload generation, failure excerpts, and deterministic analysis. `scripts/analyze_paper.py` regenerates the tables and workload plot directly from complete retained records. Its checks reject inconsistent medians and malformed timing samples. A source archive preserves the first protocol's implementation separately from follow-up lifetime changes.

The local client does not download checkpoints or corpora. Reproduction requires a Modal workspace and sufficient access to the referenced model repositories. Execute one checkpoint application at a time with at most one GPU container. The default research worker cap is one. Five rounds are planned; failures should be retained as outcomes rather than excluded without explanation. The v3 optimized-native, compiler, direct-readout, and serving study lives in a separate result namespace. `scripts/analyze_strengthened.py` checks offered-load accounting and rejects worker replacement within a serving round before generating its tables. Commit-named source snapshots and a serialized job journal identify the implementation for every v3 job.

The LaTeX source is built with `latexmk -pdf main.tex`. A CPU-only Modal typesetting entry point is supplied when TeX is unavailable locally. The review artifact includes the source and generated figures; arXiv submission has not been performed.

# B Checkpoint and Dataset Identity

Full checkpoint SHA revisions are in `src/dfip/models.py`. The typed corpus revision is `c76749ec58bd8c3d2ea706b31c333a9059c38f90`; the BANKING77 revision is `8eed2b348ec07d40ffbe91dcc4718e2f0f977714`. Data checksums are verified by remote loaders, and the derived records use random seed 20261012. Per-container package manifests must be consulted rather than assuming all checkpoints share one environment.

# C External Serving Accounting

# D Initial Fallback-Image Resident Results

Table 6 reports the short Boolean workload, while Figure 3 covers all nine shapes. These are experimental v1 policies in the initial image, kept separately from the main optimized-native comparison. The main pattern is that repeated short requests benefit more than long, batched, or multi-question paths. The tested policy deliberately falls back when capture is unsupported; different internal paths also explain why a single model-wide speedup is misleading.

The native baselines themselves vary across containers. Pairing measurements avoids dividing unrelated cloud runs, but it does not eliminate within-round mode-order effects, thermal or clock changes, or software allocator history. The Security-One native-before/native-after comparison is particularly useful for avoiding attribution of such variation to an output-head patch. These results describe resident serial-call latency; reciprocal latency is a serial service-rate estimate, not measured concurrent serving throughput.

# E Separate Follow-up Experiments

The synchronized-retirement follow-up uses source commit `b8f028d57a499e29736585675b13b02a034351af` and a separate result namespace. Its GPU kernel checks pass, including 60 changed-shape calls with a four-entry cache, eviction, and clearing. Julia still fails during the derived-workload run; synchronization alone does not establish model-level robustness.

An exploratory Clef container with `causal-conv1d` 1.7.0 installed completes all workloads and fidelity checks. Its short Boolean native median is 60.31 ms and its graphed median is 37.10 ms, a paired 1.63× ratio. This is one container, without a confidence interval, and is not pooled into Table 6. It does not establish a causal comparison between extension installation and the earlier fallback image, since environments and containers differ.

The typed fidelity set has no label or probability changes in that follow-up. On the 1,024 derived BANKING77 decisions, the graph path has zero winning-label flips but maximum probability drift 0.0129 and maximum expected-score drift 0.0189. One probability slot crosses 0.1 and another crosses 0.25. The exact IDs and probabilities are archived alongside native and graphed predictions. The 120-call changing-shape trace preserves labels with maximum probability drift 0.0012. These findings strengthen the distinction between decision agreement and probability fidelity; they do not justify enabling the experimental path by default.

| Checkpoint | Mode | Offered/s | $n$ | Completed/s | p95 (ms) | Loss (%) |
|---|---|---|---|---|---|---|
| Clef-Flash | native | 5 | 3 | 5.0 | 335 | 0.0 |
| Clef-Flash | native | 15 | 3 | 12.2 | 10078 | 4.5 |
| Clef-Flash | native | 30 | 3 | 11.6 | 12398 | 54.3 |
| Clef-Flash | graph | 5 | 3 | 2.7 | 39440 | 4.9 |
| Clef-Flash | graph | 15 | 3 | 3.5 | 41558 | 62.6 |
| Clef-Flash | graph | 30 | 3 | 2.9 | 44330 | 83.1 |
| PPLX-Decider | native | 5 | 3 | 5.0 | 810 | 0.0 |
| PPLX-Decider | native | 15 | 3 | 10.0 | 6998 | 23.4 |
| PPLX-Decider | native | 30 | 3 | 10.4 | 13393 | 58.3 |
| PPLX-Decider | graph | 5 | 3 | 2.0 | 49415 | 23.0 |
| PPLX-Decider | graph | 15 | 2 | 3.4 | 47046 | 63.4 |
| PPLX-Decider | graph | 30 | 2 | 2.9 | 46422 | 83.1 |
| Security-One | native | 5 | 3 | 5.0 | 448 | 0.0 |
| Security-One | native | 15 | 3 | 14.9 | 943 | 0.0 |
| Security-One | native | 30 | 3 | 18.3 | 6950 | 31.8 |
| Security-One | parallel | 5 | 3 | 5.0 | 472 | 0.0 |
| Security-One | parallel | 15 | 3 | 14.8 | 1012 | 0.0 |
| Security-One | parallel | 30 | 3 | 29.8 | 647 | 0.0 |

Table 5: External v3 traffic. $n$ counts observed container phases at that rate, including phases from failed rounds. Completed/s counts responses within the 60-second arrival window; later completions are retained separately. p95 is the geometric mean of container quantiles for successful requests. Loss includes drops and errors. Graph denotes experimental replay; parallel denotes concurrent dispatch to native SGLang. Raw records include p99, tail ranges, drain time, client dispatch delay, and missing-request accounting.

| Checkpoint | Complete rounds | Native median (ms) | Tested median (ms) | Paired speedup [95% CI] |
|---|---|---|---|---|
| Julia-1 | 0 | – | – | GPU error |
| d1-3B | 3 | 18.69 | 10.68 | 1.75 [1.26, 2.07] |
| Clef-Flash | 3 | 61.06 | 37.77 | 1.62 [1.60, 1.66] |
| PPLX-Decider | 5 | 95.55 | 34.23 | 2.79 [2.17, 3.23] |
| Security-One | 5 | 37.94 | 38.74 | 0.98 [0.90, 1.04] |

Table 6: Short Boolean request. Latencies are geometric means of container-level medians. Liquid/Clef intervals condition on three of five planned rounds. Security-One tests the experimental Triton head against bracketing native SGLang; the selected deployed path remains native. Julia's initial expanded protocol failed rather than yielding a repeated-run estimate.

| Checkpoint | Decisions | Flips | Crossings | Max $\lvert\Delta p\rvert$ |
|---|---|---|---|---|
| Clef-Flash | 3024 | 0 | 0 | 0.0062 |
| PPLX-Decider | 3024 | 0 | 0 | 0 |
| d1-3B | 3024 | 0 | 0 | 0.00738 |
| Security-One | 3024 | 0 | 0 | 1.82e-07 |

Table 7: Returned-distribution fidelity for retained complete round zero: 2,000 typed questions plus 1,024 derived BANKING77 decisions per checkpoint. Crossings count probability slots over all five thresholds. Security-One uses the experimental Triton head. This is empirical agreement, not an all-input guarantee.

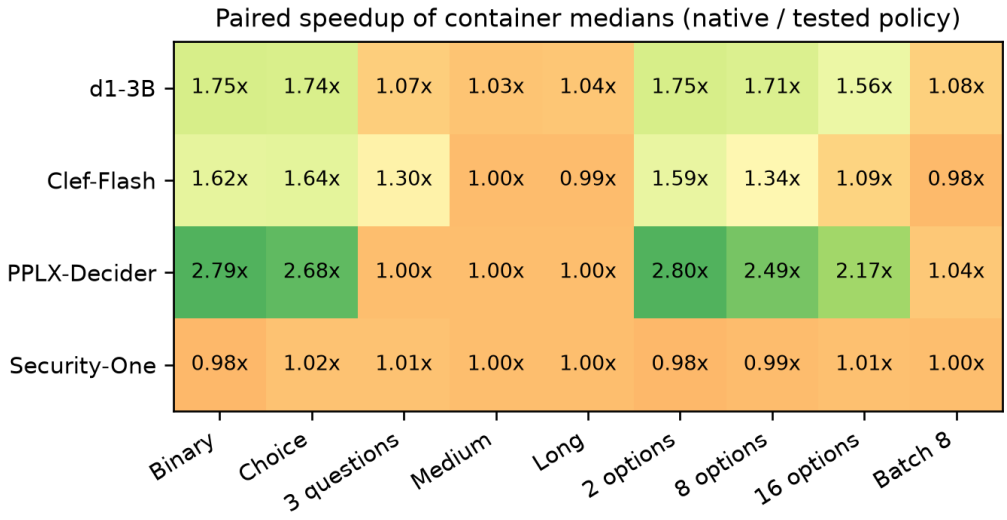



Figure 3: Paired resident-API speedup across workloads. Values below 1 indicate a slowdown. Security-One is a head experiment, not the selected deployed policy. The plot excludes Julia because its initial expanded run failed. Per-round samples and conditional intervals for every cell are in the artifact.